

Note that for the [CTAttached.SNK](#) state, there are further limitations on maximum current (see Section 4.6.2.5).

4.5.2.3.1 PowerDefault.SNK Sub-State

This sub-state supports Sinks consuming current within the lowest range (default) of Source-supplied current.

4.5.2.3.1.1 PowerDefault.SNK Requirements

The port shall draw no more than the default USB power from VBUS. See Section 4.6.2.1.

If the port wants to consume more than the default USB power, it shall monitor [vRd](#) to determine if more current is available from the Source.

4.5.2.3.1.2 Exiting from PowerDefault.SNK

For any change in [vRd](#) indicating a change in allowable power, the port shall not transition until the new [vRd](#) has been stable for at least [tRpValueChange](#).

For a [vRd](#) in the [vRd-1.5](#) range, the port shall transition to the [Power1.5.SNK Sub-State](#).

For a [vRd](#) in the [vRd-3.0](#) range, the port shall transition to the [Power3.0.SNK Sub-State](#).

4.5.2.3.2 Power1.5.SNK Sub-State

This sub-state supports Sinks consuming current within the two lower ranges (default and 1.5 A) of Source-supplied current.

4.5.2.3.2.1 Power1.5.SNK Requirements

The port shall draw no more than 1.5 A from VBUS.

The port shall monitor [vRd](#) while it is in this sub-state.

4.5.2.3.2.2 Exiting from Power1.5.SNK

For any change in [vRd](#) indicating a change in allowable power, the port shall not transition until the new [vRd](#) has been stable for at least [tRpValueChange](#).

For a [vRd](#) in the [vRd-USB](#) range, the port shall transition to the [PowerDefault.SNK Sub-State](#) and reduce its power consumption to the new range within [tSinkAdj](#).

For a [vRd](#) in the [vRd-3.0](#) range, the port shall transition to the [Power3.0.SNK Sub-State](#).

4.5.2.3.3 Power3.0.SNK Sub-State

This sub-state supports Sinks consuming current within all three ranges (default, 1.5 A and 3.0 A) of Source-supplied current.

4.5.2.3.3.1 Power3.0.SNK Requirements

The port shall draw no more than 3.0 A from VBUS.

The port shall monitor [vRd](#) while it is in this sub-state.

4.5.2.3.3.2 Exiting from Power3.0.SNK

For any change in [vRd](#) indicating a change in allowable power, the port shall not transition until the new [vRd](#) has been stable for at least [tRpValueChange](#).

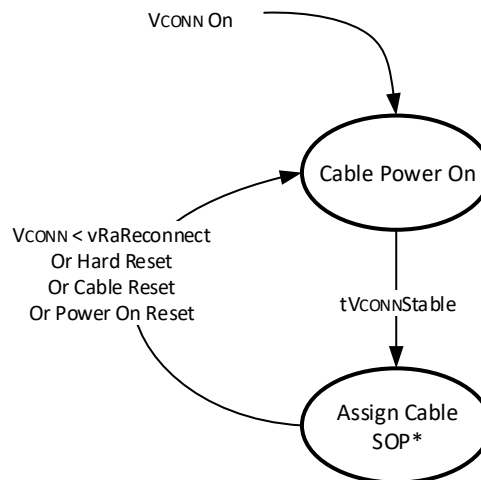
For a [vRd](#) in the [vRd-USB](#) range, the port shall transition to the [PowerDefault.SNK Sub-State](#) and reduce its power consumption to the new range within [tSinkAdj](#).

For a [vRd](#) in the [vRd-1.5](#) range, the port shall transition to the [Power1.5.SNK Sub-State](#) and reduce its power consumption to the new range within [tSinkAdj](#).

4.5.2.4 Cable State Machine Requirements

Figure 4-20 illustrates the cable eMarker connection state diagram.

Figure 4-20 Cable eMarker State Diagram



4.5.2.4.1 Cable Power On State

This state appears in Figure 4-20. This is the initial power on state for each eMarker in the cable when VCONN is applied.

4.5.2.4.1.1 Cable Power On State Requirements

Each eMarker in the cable shall present [Ra](#) when no VCONN is applied.

Each eMarker in the cable shall power on and may continue to present [Ra](#) in this state.

The cable shall not respond to SOP' and SOP'' commands in this state.

4.5.2.4.1.2 Exiting from Cable Power On State

Each eMarker in a passive or active cable shall transition to Assign Cable SOP* when it has completed its boot process. Each eMarker shall transition to Assign Cable SOP* within [tVCONNStable](#).

4.5.2.4.2 Assign Cable SOP* State

This state appears in Figure 4-20.

Typically, a passive cable has only one eMarker powered at a time. This cable eMarker in a passive cable shall respond to SOP' in this state. If two eMarkers are powered at the same time in a passive cable, then one shall respond to a pre-set SOP' and the other to SOP''.

Each cable eMarker in an active cable shall respond to a pre-set SOP' or SOP''. If only one eMarker exists in the cable, it shall only respond to SOP'.

Table 4-31 CC Timing

	Minimum	Maximum	Description
tCCDebounce	100 ms	200 ms	Time a port shall wait before it can determine it is attached
tPDDebounce	10 ms	20 ms	Time a Sink port shall wait before it can determine it is detached due to the potential for USB PD signaling on CC as described in the state definitions.
tTryCCDebounce	10 ms	20 ms	Time a port shall wait before it can determine it is re-attached during the try-wait process.
tErrorRecovery	25 ms		Time a self-powered port shall remain in the ErrorRecovery state.
	240 ms		Time a source shall remain in the ErrorRecovery state if it was sourcing VCONN in the previous state.
tRpValueChange	10 ms	20 ms	Time a Sink port shall wait before it can determine there has been a change in Rp where CC is not BMC Idle or the port is unable to detect BMC Idle.
	0 ms	5 ms	Time a Sink port shall wait before it can determine that there has been a change in Rp when USB PD signaling can be detected by the port and CC line is BMC Idle.
tSRCDisconnect	0 ms	20 ms	Time a Source shall detect the SRC.Open state. The Source should detect the SRC.Open state as quickly as practical.
tNoToggleConnect	0 ms	5 ms	Time to detect connection when neither Port Partner is toggling.
tOnePortToggleConnect	0 ms	80 ms	Time to detect connection when one Port Partner is toggling (0ms ... dcSRC.DRP max * tDRP max + 2 * tNoToggleConnect).
tTwoPortToggleConnect	0 ms	510 ms	Time to detect connection when both Port Partners are toggling (0ms ... 5 * tDRP max + 2 * tNoToggleConnect).

	Minimum	Maximum	Description
tVPDCTDD	30 μ s	5 ms	Time for a Charge-Through VCONN-Powered USB Device to detect that the Charge-Through source has disconnected from CC after VBUS has been removed, transition to CTUnattached.VPD , and re-apply its Rp termination advertising 3.0 A on the host port CC.
tVPDDisable	25 ms		Time for a Charge-Through VCONN-Powered USB Device shall remain in CTDisabled.VPD state.

4.11.3 Voltage Parameters

Table 4-32, Table 4-33 and Table 4-34 provide the CC voltage values that a Source shall use to detect what is attached based on the [USB Type-C Current](#) advertisement (Default USB, 1.5 A @ 5 V, or 3.0 A @ 5 V) that the Source is offering.

Table 4-32 CC Voltages on Source Side – Default USB

	Minimum Voltage	Maximum Voltage	Threshold
Powered cable/adapter (vRa)	0.00 V	0.15 V	0.20 V
Sink (vRd)	0.25 V	1.50 V	1.60 V
No connect (vOPEN)	1.65 V		

Table 4-33 CC Voltages on Source Side – 1.5 A @ 5 V

	Minimum Voltage	Maximum Voltage	Threshold
Powered cable/adapter (vRa)	0.00 V	0.35 V	0.40 V
Sink (vRd)	0.45 V	1.50 V	1.60 V
No connect (vOPEN)	1.65 V		

Table 4-34 CC Voltages on Source Side – 3.0 A @ 5 V

	Minimum Voltage	Maximum Voltage	Threshold
Powered cable/adaptor (vRa)	0.00 V	0.75 V	0.80 V
Sink (vRd)	0.85 V	2.45 V	2.60 V
No connect (vOPEN)	2.75 V		

Table 4-35 provides the CC voltage values that shall be detected across a Sink's [Rd](#) for a Sink that does not support higher than default [USB Type-C Current](#) Source advertisements.

Table 4-35 Voltage on Sink CC Pins (Default USB Type-C Current only)

Detection	Min voltage	Max voltage	Threshold
vRa	–0.25 V	0.15 V	0.2 V
vRd-Connect	0.25 V	2.18 V	

Table 4-36 provides the CC voltage values that shall be detected across a Sink's [Rd](#) for a Sink that implements detection of higher than default [USB Type-C Current](#) Source advertisements. This table includes consideration for the effect that the IR drop across the cable GND has on the voltage across the Sink's [Rd](#).

Table 4-36 Voltage on Sink CC pins (Multiple Source Current Advertisements)

Detection	Min voltage	Max voltage	Threshold
vRa	–0.25 V	0.15 V	0.2 V
vRd-Connect	0.25 V	2.04 V	
vRd-USB	0.25 V	0.61 V	0.66 V
vRd-1.5	0.70 V	1.16 V	1.23 V
vRd-3.0	1.31 V	2.04 V	